Tim Cunningham

Assistant Professor
Department of Physics
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Assistant Professor and STFC Ernest Rutherford Fellow in the Department of Physics at the University of Warwick, formerly NASA Hubble Fellow at Harvard & Smithsonian Center for Astrophysics, utilizing multi-dimensional radiation-hydrodynamic simulations and multi-wavelength observational strategies to study convection in degenerate stars and white dwarf populations. Experienced observer with spectroscopic and photometric expertise. Commitment to leadership roles, mentoring, outreach and teaching.

Academic Qualifications

University of Warwick — **Ph.D. in Physics**
Coventry, UK — *September 2020*

Group:	*Astronomy and Astrophysics*
Thesis:	Convective overshoot in the atmosphere of white dwarf stars
Supervisor:	Pier-Emmanuel Tremblay

King's College London — **MSci in Physics**
London, UK — *September 2014*

Reading:	Physics
Class:	First Class Honors

Professional Appointments

University of Warwick — **Assistant Professor**
Coventry, UK - STFC Ernest Rutherford Fellow — *Sept 2026–*

Harvard University — **NASA Hubble Fellow**
Cambridge, MA, US — *Sept 2023–Aug 2026*

University of Warwick — **Postdoctoral Research Fellow**
Coventry, UK — *July 2020 – Sept 2023*

University of California, Santa Barbara — **Research Assistant**
Santa Barbara, CA, US — *Apr – Sept 2013*

Publications (with links)

Metrics: h-index=21 on 45+ abstracts with 2,000+ citations (Google Scholar; ORCID)

As Lead Author

10. **Cunningham, T.**, Caiazzo, I, Raymond, J., et al. 2026, **ApJ**, accepted — *2026*
    Detection of knots in the supernova type Iax remnant Pa 30
9. **Cunningham, T.**, Caiazzo, I, Sienkiewicz, G., et al. 2025b, **MNRAS**, 540, 633 — *2025*
    Serendipitous discovery of two new polars evolved past the period bounce
8. **Cunningham, T.**, Tremblay, P.-E., O'Brien, M., et al. 2025a, **MNRAS**, 539, 2021. — *2025*
    The dearth of high-mass hydrogen-atmosphere metal-polluted white dwarfs within 40 pc
7. **Cunningham, T.**, Caiazzo, I., Prusinski, N., et al. 2024b, **ApJL**, 975, L7 — *2024*
    Expansion properties of the young supernova type Iax remnant Pa 30 revealed

6. **Cunningham, T.**, Tremblay, P.-E., O'Brien, M. 2024a, **MNRAS**, 527, 3602 *2024*
Initial-final mass relation from white dwarfs within 40 pc

5. **Cunningham, T.**, Wheatley, P. J., Tremblay, P.-E., et al. 2022, **Nature**, 602, 219 *2022*
A White Dwarf Accreting Planetary Material Determined from X-ray Observations

4. **Cunningham, T.**, Tremblay, P.-E., Bauer, E., et al. 2021, **MNRAS**, 503, 1646 *2021*
Horizontal Spreading of Planetary Debris Accreted by White Dwarfs

3. **Cunningham, T.**, Tremblay, P.-E., Gentile Fusillo, N. P., et al. 2020, **MNRAS**, 492, 3540 *2020*
From Hydrogen to Helium: The Spectral Evolution of White Dwarfs as Evidence for Convective Mixing

2. **Cunningham, T.**, Tremblay, P.-E., Freytag, B., et al. 2019, **MNRAS**, 488, 2503 *2019*
Convective Overshoot and Macroscopic Diffusion in Pure-Hydrogen-Atmosphere White Dwarfs

1. **Cunningham T.**, Wolf W. M. & Bildsten L. 2015, **ApJ**, 803, 76 *2015*
Photoionization Heating of Nova Ejecta by the Post-Outburst Supersoft Source

As Co-Author

39. Cherubim, C., et al. 2026, **ApJ**, in prep. *2026*
Variable X-ray flux may explain variable helium escape from the habitable-zone exoplanet LHS 1140b

38. Ould Rouis, L. B., et al. 2026, **ApJ**, in prep. *2026*
Using White Dwarf Gaia Spectra to Increase the Sample of Massive Remnant Planetary System Hosts

37. van Roestel, J., et al. 2026, **A&A**, submitted. *2026*
Now you see me, and now you don't...Obscuration of white dwarfs by dust and debris as the result of planetesimal disruption events

36. O'Brien, M., et al. 2026, **MNRAS**, accepted. *2026*
White dwarfs within 13 pc: Insights from ultraviolet spectroscopy

35. Chen, W., et al. 2026, **MNRAS**, accepted. *2026*
3D Model Spectra of White Dwarfs Accreting Planetary Debris

34. Riva, F., et al. 2026, **A&A Letters**, accepted. *2026*
Kilogauss magnetic fields from simulations of small-scale dynamo action in the convective envelope of a white dwarf

33. Williams, J. T., et al. 2026, **Nature Astron.**, accepted *2026*
A white dwarf accreting a candidate second-generation planet

32. Yang, H., et al. 2026, **ApJ**, accepted. *2026*
A Minute-Cadence Deep Bulge Survey: First Data Release of DREAMS

31. Rogers, L., et al. 2026, **ApJ**, accepted. *2026*
Spectroscopic Monitoring of Metal Lines in Polluted White Dwarfs

30. Pham, D., et al. 2026, **ApJ**, accepted. *2026*
The Effects of Magnetic Accretion on the Spatial Extent of White Dwarf Pollution

29. Cherubim, C., et al. 2026, **Science**, in press. *2026*
Escaping helium atmosphere on a terrestrial planet in a nearby habitable zone

28. Desai, A., et al. 2026, **A&A**, accepted. *2026*
Circumstellar interaction in the extreme white dwarf merger remnant ZTF J1901+1458: A new class of

white dwarf merger remnants with X-ray emission

27. Elms, A., et al. 2026, **MNRAS**, 581, 1 *2026*

Detection of a weak magnetic field in the Balmer emission line white dwarf WDJ1653-1001

26. Pallathadka, A., et al. 2026, **ApJ**, 999, 251 *2025*

Double White Dwarf Binaries in SDSS-V DR19: The discovery of a rare DA+DQ white dwarf binary with 31 hour orbital period

25. Bauer, E., et al. 2026, **ApJSS**, 283, 41 *2025*

MESA Isochrones and Stellar Tracks (MIST) III. The White Dwarf Cooling Sequence

24. Cristea, A., et al. 2026, **A&A**, 706, 188 *2026*

A half-ring of ionized circumstellar material trapped in the magnetosphere of a white dwarf merger remnant. A new class of white dwarf merger remnants with X-ray emission

23. Buchan, A., et al. 2025, **MNRAS**, 544, 2098 *2025*

Exogeological inferences from white dwarf pollutants: the impact of stellar physics

22. Kollmeier, J., et al. 2025, **AJ**, 171, 1 *2025*

Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy

21. Munday, J., et al. 2025, **MNRAS**, 541, 3494 *2025*

The DBL Survey II: towards a mass-period distribution of double white dwarf binaries

20. Roberts, E., et al. 2025, **MNRAS**, 538, 2548 *2025*

Comparison of methods used to derive the Galactic star formation history from white dwarf samples

19. Munday, J., et al. 2025, **Nature Astron.**, 9, 872 *2025*

There is a super-Chandrasekhar mass type Ia supernova progenitor located 49 pc away that will detonate in 23 Gyr

18. Sahu, S., et al. 2025, **Nature Astron.**, 9, 1347 *2025*

Ultraviolet carbon detection reveals a hot white dwarf merger remnant

17. Williams, J. T., et al. 2024, **A&A**, 691, A352 *2024*

PEWDD: A database of white dwarfs enriched by exo-planetary material

16. Elms, A. et al. 2024, **MNRAS**, 534, 2758 *2024*

A network of cooler white dwarfs as infrared standards for flux calibration

15. Munday, J. et al. 2024, **MNRAS**, 532, 2534 *2024*

The DBL Survey I: discovery of 34 double-lined double white dwarf binaries

14. O'Brien, M. W., et al. 2023, **MNRAS**, 527, 8687 *2024*

The 40 parsec sample of white dwarfs from Gaia

13. Corrales, L., et al. 2023, (AXIS white paper) *2023*

The life cycle of stars and their planets from the high energy perspective

12. Rebassa-Mansergas, A., et al. 2023, **MNRAS**, 526, 4787 *2023*

Main-sequence companions to white dwarfs – II. The age–activity–rotation relation from a sample of Gaia common proper motion pairs

11. O'Brien, M. W., et al. 2023, **MNRAS**, 518, 3055 *2023*

Gaia white dwarfs within 40 pc III: spectroscopic observations of new candidates in the southern hemisphere

10. Elms, A. K., et al. 2022, **MNRAS**, 517, 4557 *2022*
Spectral analysis of ultra-cool white dwarfs polluted by planetary debris

9. Raddi, R., et al. 2022, **A&A**, 658, A22 *2022*
Kinematic properties of white dwarfs. Galactic orbital parameters and age-velocity dispersion relation

8. Rebassa-Mansergas, A., et al. 2021, **MNRAS**, 505, 3165 *2021*
Constraining the solar neighbourhood age-metallicity relation from white dwarf-main sequence binaries

7. Veras, D., et al. 2021, **MNRAS**, 506, 1148 *2021*
The entry geometry and velocity of planetary debris into the Roche sphere of a white dwarf

6. McCleery, J., et al. 2020, **MNRAS**, 499, 1890 *2020*
Gaia white dwarfs within 40 pc II: the volume-limited Northern hemisphere sample

5. Tremblay, P.-E., et al. 2020, **MNRAS**, 497, 130 *2020*
Gaia White Dwarfs within 40 pc I: Spectroscopic Observations of New Candidates

4. Green, M. J., et al. 2020, **MNRAS**, 496, 1243 *2020*
Spectroscopic and Photometric Periods of Six Ultracompact Accreting Binaries

3. Tremblay, P.-E., et al. 2019, **MNRAS**, 482, 5222 *2019*
Fundamental Parameter Accuracy of DA and DB White Dwarfs in Gaia Data Release 2

2. Gentile Fusillo, N. P., et al. 2019, **MNRAS**, 482, 4570 *2019*
A Gaia Data Release 2 catalogue of white dwarfs and a comparison with SDSS

1. Tremblay, P.-E., et al. 2019, **Nature**, 565, 202 *2019*
Core Crystallization and Pile-up in the Cooling Sequence of Evolving White Dwarfs

Telescope Time Awarded as PI

Metrics: 1.1 Ms (315 hours) of *Chandra*/*XMM-Newton* time, 44 hours of *JWST* time, 22 orbits of *HST* time, and 29 nights of >6.5m ground-based spectroscopy and photometry awarded as PI:

Space-based

XMM-Newton: AO 25 (#098376) – 15 hours (54 ks, priority A) *2025*
"X-ray emission from white dwarfs accreting planetary debris"

XMM-Newton: AO 25 (#098376) – 36 hours (128 ks, priority C) *2025*
"Metal-polluted white dwarfs as novel X-ray sources"

Chandra+JWST: Cycle 27 (#27300224) – 14 hours (50 ks) with ACIS-S *2025*
\+ 5.5 hours JWST NIRSec/MIRI
"Accretion-induced X-ray emission from the closest candidate polar"
PI: Cunningham

HST: Cycle 33 (GO #18082) – 16 orbits *2025*
"A Deeper Look at the Type Iax Supernova Remnant Pa 30"
PI: Cunningham

HST: Cycle 33–35 (GO #18078) – 6 orbits *2025*
"Constraining variability of white dwarf planetary debris accretion"
PI: Cunningham

JWST: Cycle 4 (GO #9077) – 11.8 hours *2025*

"The iconic debris disk surrounding the white dwarf G29-38"
PI: Cunningham

JWST: Cycle 4 (GO #9111) – 26.7 hours *2025*
"An Infrared View of the Type Iax Supernova Remnant Pa 30"
Co-PI: Cunningham | PI: Caiazzo

XMM-Newton: AO 24 (#096498) – 36 hours (128 ks, priority C) *2024*
"Metal-polluted white dwarfs as novel X-ray sources"

XMM-Newton: AO 23 (#09418801) – 34 hours (124 ks, priority C) *2023*
"Metal-polluted white dwarfs as novel X-ray sources"

Chandra: Cycle 25 (#25200153) – 104 hours (375 ks) with HRC-I *2023*
"Constraining the time-dependent accretion rate of a novel class of X-ray source"

Chandra: Cycle 25 (#25200459) – 10 hours (35 ks) with ACIS-S *2023*
"Earliest X-rays from a possible late thermal pulse?"

Chandra: Cycle 24 (#24200446) – 35 hours (125 ks) with HRC-I *2022*
"Constraining the time-dependent accretion rate of a novel class of X-ray source"

Chandra: Cycle 22 (#22200039) – 32 hours (115 ks) with ACIS-S *2020*
"First X-rays from white dwarf planet accretion: a definitive test of convective overshoot"

Ground-based

GTC: 2026B – 19.5 hours high-cadence optical multi-band photometry with HiPERCAM *2026*
"From the NUV to the MIR: A Panchromatic View of G29-38"

IRTF: 2026B – 33 hours optical and near-infrared spectroscopy with SpeX *2026*
"Debris disk response to the stellar pulsations of G29-38"

Gemini: 2026A – 6.4 hours optical spectroscopy with GMOS-S *2026*
"Phase resolved spectroscopy for Ultramassive WDs"

Gemini: 2026A – 5.0 hours optical spectroscopy with GMOS-N *2025*
"Phase resolved spectroscopy for Ultramassive WDs"

Gemini: Fast Turnaround May 2025 – 6.9 hours imaging with GMOS-N *2025*
"A Deeper Look at the Type Iax Supernova Remnant Pa 30"

FLWO: Semester 2025B – 16 nights with FLWO/KeplerCam *2025*
"JWST synergistic optical monitoring of the iconic debris disk surrounding the white dwarf G29-38"

MMT: 2025B – 3 nights spectroscopy with Binospec *2025*
"Double-lined double white dwarf binaries as progenitors for Type Ia supernovae"

MMT: 2025B – 1 night imaging with MMTCam *2025*
"A Deeper Look at the Type Iax Supernova Remnant Pa 30"

MMT: 2025B – 2 nights imaging with MMIRS *2025*
"Novel accreting white dwarfs identified through X-ray emission and photometric variability"

Magellan: Semester 2025A – 4.5 nights with FourStar/MagE *2025*
"Novel isolated white dwarf X-ray sources from XMM-Newton"

Magellan: Semester 2025A – 3 nights with FourStar/MagE *2025*

"Novel isolated white dwarf X-ray sources from XMM-Newton"

MMT: Semester 2025A – 3 nights with Binospec *2025*
"Novel isolated white dwarf X-ray sources from XMM-Newton"

MMT: Semester 2024B – 1 nights with Binospec *2024*
"White dwarf gaseous disks discovered in SDSS-V"

MMT: Semester 2024B – 1 nights with Binospec *2024*
"Extreme white dwarfs in the Solar neighbourhood"

MMT: Semester 2024B – 1 nights with MMIRS *2024*
"Confirming the first strongly asynchronous polar"

MMT: Semester 2024B – 2 nights with Binospec *2024*
"Novel isolated white dwarf X-ray sources from XMM-Newton"

Magellan: Semester 2024B – 4 nights with FourStar/FIRE/MagE *2024*
"Novel isolated white dwarf X-ray sources from XMM-Newton"

Magellan: Semester 2024A – 1 night with FourStar *2024*
"Infrared time-series of the oldest accreting magnetic white"

Magellan: Semester 2024A – 2 nights with MagE *2024*
"Novel isolated white dwarf X-ray sources from XMM-Newton"

Grants awarded

Metrics:
- $2,345,000 in independent fellowship grants offered.
- $659,000 of observing grants awarded for *Chandra*/*HST*/*JWST* programs as PI/Co-PI.

Grants offered for independent fellowships:

STFC Ernest Rutherford Fellowship, University of Warwick [future position] *Sept 2026–31*
Awarded grant value: £777,384 ($1,000,000)

NASA Hubble Fellowship (STScI), Harvard University [current position] *Sept 2023–26*
Awarded grant value: $400,000

51 Pegasi b Fellowship (Heising-Simons), The Ohio State University [declined] *2023*
Awarded grant value: $450,000

Stephen Hawking Fellowship (EPSRC), University of Cambridge [declined] *2024*
Awarded grant value: £385,000 ($495,000)

Grants awarded for *JWST* PI time:

JWST: Cycle 4 (GO 9077) – $121,000 *2025*
Science PI: **Cunningham**, Administrative PI: Charbonneau

JWST: Cycle 4 (GO 9111) – $192,000 *2025*
Science PIs: Caiazzo/**Cunningham**, Administrative PI: Charbonneau

Grants awarded for *HST* Co-PI time:

HST: Cycle 32 (HST-GO-17720) – $69,000 *2024*
Science PI: Caiazzo, US Science PI: **Cunningham**, Administrative PI: Charbonneau

Grants awarded for *Chandra* PI time:

Chandra: Cycle 27 (#27300224) – $62,000 *2025*

Science PI: **Cunningham**, Administrative PI: Hermes

Chandra: Cycle 25 (#25200153) – $120,000 *2023*

Science PI: **Cunningham**, Administrative PI: Charbonneau

Chandra: Cycle 25 (#25200459) – $60,000 *2023*

Science PI: **Cunningham**, Administrative PI: Charbonneau

Chandra: Cycle 24 (#24200446) – $35,000 *2022*

Science PI: **Cunningham**, Administrative PI: Corrales

Professional Service

Collaborations: *2022–*

- ***NewAthena* - ESA proposed Flagship X-ray Mission**:
  - Stars Science Working Group Member
  - Compact Objects Science Working Group Member
  - (study phase - adoption decision in 2027)
- ***AXIS* - NASA Probe Mission Concept**:
  - Stars and Exoplanets Science Working Group Member
  - (selected for Phase A funding in October 2024)
- **Zwicky Transient Facility**:
  - External collaborator - Stars Group
- **4MOST**:
  - Team Member of the White Dwarf Binary (WDB) Survey (PI: Toloza)
- **SDSS-V**:
  - Membership through current institution

Journal referee: *2021–*

- Astronomy & Astrophysics (A&A)
- Monthly Notices of the Royal Astronomical Society (MNRAS)
- Monthly Notices of the Royal Astronomical Society (MNRAS) Letters
- Astrophysical Journal (ApJ)
- Astrophysical Journal Letters (ApJL)

Peer Review & Time Allocation Committee Panels: *2023–*

- ***Hubble Space Telescope***
- ***Chandra X-ray Observatory***:
  - Deputy Chair
- **NASA Keck**
  - Panel chair
- **FINESST**:
  - External Reviewer
- **NASA ROSES**:
  - Astrophysics Theory Program (ATP)
  - *ULTRASAT* – Ultraviolet Transient Astronomy Satellite Participating Scientist

Scientific Organising Committee (SOC) – Magellan Science Meeting *2025*

- EPL Carnegie, 5–7 May 2025

Session organiser – National Astronomy Meeting 2023, Cardiff, 3–7 July *2023*

- *"Open questions in stellar (radiation-) fluid dynamics: from dwarfs to giants"*

Session organiser – National Astronomy Meeting 2022, Warwick, 11–15 July *2022*

- *"Know Your Neighbour: White Dwarf Researchers in the UK"*
- *"3D Hydrodynamics and Magnetohydrodynamics: Stars and their Environment"*

Scientific Organising Committee (SOC) – Physics Days at Warwick, Warwick, 5/6 July *2021*

- *"Summer workshop on white dwarfs and related objects"*

Session chair – National Astronomy Meeting 2019, Lancaster, 30 June–4 July *2019*

- *"Current Developments in Numerical Astrophysics"*

Contributed Presentations

Talks

EuroWD24 **Vienna, AT**
24th European workshop on white dwarfs – [*talk*] *August 2026*

Debris Disk Connections **Cambridge, UK**
Debris Disk Connections Workshop – [*talk*] *July 2026*

TASC10 **Aarhus, DK**
TASC10/KASC17 – [*talk - awarded second-best talk*] *July 2026*

Exo VI **Porto, PT**
Exoplanets VI – [*talk*] *July 2026*

Hubble & Webb (VIII) **Vienna, AT**
Enriching the Universe: ... – [*talk*] *April 2026*

Center for Computational Astrophysics **New York, US**
Stars & Plasma Astrophysics Group – [*talk*] *March 2026*

Center for Astrophysics **Cambridge, MA, US**
pH Lecture – [*endowed colloquium*] *February 2026*

Yale University **New Haven, CT, US**
Seminar – [*seminar*] *January 2026*

Boston University **Boston, MA, US**
Colloquium – [*colloquium*] *October 2025*

NHFP Symposium - STScI **Baltimore, US**
NASA Hubble Fellowship Program Symposium – [*talk*] *October 2025*

Purdue University **Indiana, US**
Astrophysics seminar – [*seminar*] *September 2025*

University of Michigan **Ann Arbor, MI, US**
Astrophysics seminar – [*seminar*] *September 2025*

Harvard Origins of Life Initiative **Cambridge, MA, US**
Chalk talk – [*talk*] *March 2025*

AAS Meeting **Maryland, US**
245th AAS Meeting – [*talk*] *January 2025*

AXIS Seminar | **Online Seminar**
AXIS Science Team Seminar – [*talk*] | *October 2024*

NHFP Symposium - Caltech | **Pasadena, CA, US**
NASA Hubble Fellowship Program Symposium – [*talk*] | *September 2024*

TANDEM Seminar Series | **Cambridge, MA, US**
Harvard & Smithsonian | Center for Astrophysics – [*talk*] | *August 2024*

EuroWD24 | **Barcelona, ES**
23rd European workshop on white dwarfs – [*talk*] | *July 2024*

Institute for Science and Technology Austria (ISTA) | **Vienna, AT**
Astrophysics seminar – [*seminar*] | *June 2024*

Exo V | **Leiden, NL**
Exoplanets V – [*talk, plenary session*] | *June 2024*

NORDITA | **Stockholm, SE**
Physics and Astrophysics seminar – [*seminar*] | *June 2024*

ESAC/XMM-Newton | **ESAC, Madrid, ES**
The X-ray mysteries of neutron stars and white dwarfs – [*talk*] | *June 2024*

Caltech | **Pasadena, CA, US**
Astronomy and Astrophysics colloquium – [*colloquium*] | *April 2024*

ExSS V | **Christchurch, NZ**
Extreme Solar Systems V – [*talk*] | *March 2024*

Tufts University | **Cambridge, MA, US**
Astronomy and Physics – [*colloquium*] | *March 2024*

CfA/Harvard | **Cambridge, MA, US**
ITC Luncheon – [*talk*] | *February 2024*

UCLA | **Los Angeles, CA, US**
Astrophysics seminar – [*seminar*] | *February 2024*

NHFP Symposium - Harvard University | **Cambridge, MA, US**
NASA Hubble Fellowship Program Symposium – [*talk*] | *September 2024*

University of Exeter | **Exeter, UK**
Astrophysics Group – [*seminar*] | *June 2023*

Aspen Center for Physics | **Aspen, CO**
Conference on late-stage planetary systems – [*talk*] | *April 2023*

KITP | **California, US**
White Dwarfs as Probes of the Evolution of Planets, Stars...Universe – [*talk*] | *December 2022*

CfA/Harvard | **Boston, US**
High Energy Seminars, Centre for Astrophysics/Harvard & Smithsonian – [*seminar*] | *October 2022*

University of Cambridge | **Cambridge, UK**
Exoplanet Seminar, Cavendish Astrophysics/Institute of Astronomy – [*seminar*] | *October 2022*

Armagh Observatory & Planetarium | **Armagh, NI**
Astronomy Group – [*seminar*] | *September 2022*

UKEXOM 2022 — **Edinburgh, UK**
UK Exoplanet Community Meeting – [*talk*] — *September 2022*

EUROWD22 — **Tübingen, Germany**
22nd *European Workshop on White Dwarfs* – [*talk*] — *August 2022*

Ohio State University — **Columbus, OH, US**
Exoplanet Group – [*seminar*] — *August 2022*

X-ray Astronomy UK — **Leicester, UK**
X-ray Astronomy Meeting UK – [*talk*] — *May 2022*

AASTCS 9: Exoplanets IV — **Las Vegas, NV, US**
Exoplanets IV – [*talk, plenary session*] — *May 2022*

Keele University — **Keele, UK**
Astrophysics Group Seminar — *January 2022*

Imperial College London — **London, UK**
Astrophysics Group Seminar — *December 2021*

KITP — **Santa Barbara, CA, US**
Transport in Stellar Interiors – [*panel discussion on chemical mixing*] — *November 2021*

KITP — **Santa Barbara, CA, US**
Probes of Transport in Stars – [*talk*] — *November 2021*

O-MESS — **Potsdam, DE**
Online - Meetings on Evolved Stars and Systems — *June 2021*

KITP — **Santa Barbara, CA, US**
KITP Online Reunion Conference: Exostar Redux – [*talk*] — *August 2020*

NORDITA — **Stockholm, Sweden**
(*CANCELLED*) *The Shifting Paradigm of Stellar Convection: from mixing...* — *March 2020*

MPIA — **Heidelberg, Germany**
Bergemann Stars Group Meeting — *November 2019*

KITP — **Santa Barbara, CA, US**
Bildsten Stellar Group Meeting — *November 2019*

IAUS 357 — **Hilo, Hawaii, US**
White dwarfs as probes of fundamental physics and... — *October 2019*

CCC2019 — **Newcastle, UK**
Compressible Convection Conference — *September 2019*

NAM2019 — **Lancaster, UK**
National Astronomy Meeting — *July 2019*

EuroWD21 — **Austin, TX, US**
21st European White Dwarf Workshop – [*talk*] — *July 2018*

EWASS 2018 — **Liverpool, UK**
European Week of Astronomy and Space Science — *April 2018*

Posters

Los Alamos National Laboratory (Public) — **Santa Fe, New Mexico, US**
Current Challenges in the Physics of White Dwarf Stars — *June 2017*

EuroWD16 — **Warwick, UK**
20th European White Dwarf Workshop — *July 2016*

SDOB7 — **Oxford, UK**
Seventh Meeting on Hot Subdwarfs and Related Objects — *July 2015*

Selected Press Coverage

- *Escaping helium atmosphere on a terrestrial planet in a nearby habitable zone* (Cherubim et al. 2026)
  - New York Times: Astronomers Find an Atmosphere on a Nearby Earthlike Planet
  - BBC: First atmosphere found on Earth-like planet in habitable zone of distant star
  - The Guardian: Earth-like exoplanet found to have an atmosphere
- *Super-Chandrasekhar mass SN Ia progenitor that will detonate in 23 Gyr* (Munday et al. 2025)
  - BBC: Two dwarf stars spotted that could explode in 22 billion years!
  - Reuters: Astronomers spot two white dwarfs doomed to die in a quadruple detonation
  - Cover of Nature Astronomy: A supernova progenitor within 50 parsecs
- *Expansion properties of the young supernova type Iax remnant Pa 30 revealed* (Cunningham et al. 2024)
  - CNN: A supernova first seen in 1181 is releasing glowing filaments
  - Scientific American: Scientists Spy a 'Dandelion' Supernova around a 'Zombie' Star
- *Spectral analysis of ultra-cool white dwarfs polluted by planetary debris* (Elms et al. 2022)
  - NewScientist: Planets around graveyard star were consumed before Earth was even born
  - Independent: New star system discovery may be oldest in our galaxy, scientists say
- *A white dwarf accreting planetary material determined from X-ray observations* (Cunningham et al. 2022)
  - Independent: Final moments of dead planet crashing into a white dwarf seen for the first time
  - Vice: Scientists have finally seen what happens to shredded worlds in deep space
- *Core crystallization and pile-up in the cooling sequence of evolving white dwarfs* (Tremblay et al. 2019)
  - Los Angeles Times: One day our sun will solidify into a giant crystal orb
  - NewScientist: Astronomers have seen dying stars slowly crystallise and turn solid

Observational Experience

Spectroscopy

KCWI — **Mauna Kea Observatory**
Keck — (*3 nights*) *2023-*
Program: *Observations of the type Iax supernova remnant Pa30* — (PI: Fuller/Polin)

MagE — **Las Campanas Observatory**
Magellan/Baade — (*2 nights*) *2023-*
Program: *Novel isolated white dwarf X-ray sources from XMM-Newton* — (PI: Cunningham)

FIRE — **Las Campanas Observatory**
Magellan/Baade — (*1 night*) *2024*
Program: *Novel isolated white dwarf X-ray sources from XMM-Newton* — (PI: Cunningham)

IDS — **IAC, La Palma**
Isaac Newton Telescope (INT) — (*5 nights*) *Jul 2019*
Program: *The age-metallicity relation: constraints from white dwarf-main sequence binaries* — (PI: Gaensicke)

ISIS — **IAC, La Palma**
William Herschel Telescope (WHT) — (*9 nights*) *Sep 2018*
Program: *The mass ratio distribution of close double white dwarf binary stars* — (PI: Marsh)

Photometry

FourStar — **Las Campanas Observatory, Chile**
Magellan/Baade — *(1 night) Nov 2023*

ULTRACAM — **La Silla, ESO, Chile**
New Technology Telescope (NTT) — *(6 nights) Jun 2018*

ULTRASPEC — **TNO, Chang Mai, Thailand**
Thai National Telescope (TNT) — *(8 nights) Feb 2018*

Wide Field Camera — **IAC, La Palma**
Isaac Newton Telescope (INT) — *(6 nights) Feb 2017*

Leadership, Outreach and Teaching

Current Students

Sumit Maheshwari – Master's student in Physics at University of Hamburg (2024).
- Project: *Near-infrared time-series photometric and spectroscopic studies of magnetic accreting white dwarfs*

Former Students

Will Brennom – post-Bachelors at the Harvard–Smithsonian Center for Astrophysics (2024).
- Project: *Spectroscopic observations of wide white dwarf companions to Hot Jupiter hosts*
- Now PhD student at Michigan State University.

Gracjan Sienkiewicz – second year undergraduate in Physics (2023).
- *Undergraduate Research Support Scheme*, University of Warwick
- Project: *Development of Accretion Models for White Dwarfs at Low Accretion Rates*
- Now PhD student at ISTA, Vienna.

Klemen Keršič – second year undergraduate in Physics (2021).
- *Undergraduate Research Support Scheme*, University of Warwick
- Project: *Modeling Accretion of Planetary Debris on White Dwarfs (2021)*
- Now MSci student at ETH, Zürich.

Lecturer — **University of Warwick**
Undergraduate module: Science and Music — *2017–2023*

Project supervisor — **University of Warwick**
Undergraduate module: Science and Music — *2017–2023*

Supervised projects include:
- *Data-to-sound: algorithmic sonifcation of images and data* (2023)
- *Quartet for the end of term: Olivier Messiaen's modes of limited transposition* (2022)
- *Turning sound into colour: Virtual Reality and insights into chromesthesia* (2020)
- *Scriabin's colour wheel: exploring chromesthesia with MIDI piano and C/C++* (2019)
- *Turning colour into sound: developing algorithms to 'hear' images* (2018)

Lab Demonstrator — **University of Warwick**
Undergraduate module: First Year Laboratory — *2016–2018*

Public Observatory Night — **Harvard College Observatory**
Center for Astrophysics — *February 2026*

Research Lecture — **King's College London**
First Year Astrophysics Course — *February 2018*

Astronomy Group Representative — **University of Warwick**
Postgraduate Staff-Student Liaison Committee — *2016–2018*

Athena Swan Diversity Representative — **University of Warwick**
Postgraduate Staff-Student Liaison Committee — *2016–2018*

Outreach Assistant — **University of Warwick**
Planterium — *2017–2018*

Astrophysics demonstrator — **University of Warwick**
XMaS Science Gala — *2017*

Technical Expertise

Programming Languages:	`Python` • `IDL` • `Fortran` • `C/C++` • `C#` • `bash`	
High Performance Computing:	`OpenMP` • `MPI` • `slurm` • `moab`	
Software (Scientific):	`CO5BOLD` • `MESA` • `LaTeX` • `Unity3D`	
Software (X-ray):	`SAS` • `CIAO` • `XSPEC` • `BXA`	
Software (Accredited):	Microsoft Office Specialist Excel 2016	May 2019–present
Software (Audio):	`Logic Pro X` • `Ableton Live` • `Sibelius`	

Interdisciplinary Accomplishments

Prize Winner — **University of Warwick**
Astronomy Meets the Arts — *2019*
Film: "A Night at the Isaac Newton Telescope"

Composer — **British Science Festival, Coventry**
Composition: "Immersed in soundwaves" — *2019*
A piece for String Quartet, Theremin and Bass Guitar (written for Coull Quartet)

Musicianship Teacher — **Morley College, London**
Centre for Young Musicians — *2014–2023*

Workshop Leader — **Shrewsbury Folk Festival, UK**
Tuneworks and Refolkus — *2018–2019*

Composer–Musical Director–Performing Artist — **Huddersfield, UK**
Proper Job Theatre Company — *2015–2019*